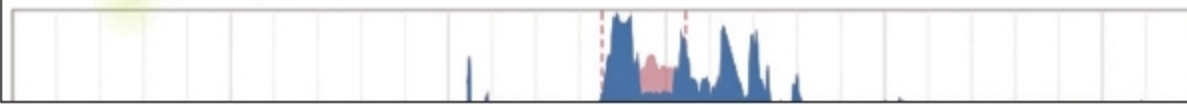


**Boot chart for sourya (Wed Feb 6 15:52:47 IST 2019)**

```
uname: Linux 3.16.0 #2 SMP Mon Feb 4 15:31:32 IST 2019 x86_64
release: Ubuntu 14.04.5 LTS
CPU: Intel(R) Core(TM) i7-7820HK CPU @ 2.90GHz model name: Intel(R) Core(TM) i7-7820HK CPU @ 2.90GHz
kernel options: BOOT_IMAGE=/boot/vmlinuz-3.16.0 root=UUID=f21b3fe3-a58e-4103-881a-ee48264b81dc ro quiet
time: 00:50 98
```

■ CPU (user+sys) ■ I/O (wait)

Figure 7: Boot speed before making changes to the *rc* file**Boot chart for sourya (Thu Feb 7 11:22:49 IST 2019)**

```
uname: Linux 3.16.0 #2 SMP Mon Feb 4 15:31:32 IST 2019 x86_64
release: Ubuntu 14.04.5 LTS
CPU: Intel(R) Core(TM) i7-7820HK CPU @ 2.90GHz model name: Intel(R) Core(TM) i7-7820HK
kernel options: BOOT_IMAGE=/boot/vmlinuz-3.16.0 root=UUID=f21b3fe3-a58e-4103-881a-ee48264b81dc ro quiet
time: 00:23.85
```

■ CPU (user+sys) ■ I/O (wait)

Figure 8: Boot speed after making changes to the *rc* file

```
sourya@sourya:~$ systemd-analyze
Startup finished in 22.320s (kernel) + 28.139s (userspace) = 50.460s
graphical.target reached after 28.114s in userspace
sourya@sourya:~$
```

Figure 9: Boot speed before using e4rat

```
sourya@sourya: ~
File Edit View Search Terminal Help
sourya@sourya:~$ systemd-analyze
Startup finished in 9.065s (kernel) + 6.220s (userspace) = 15.286s
graphical.target reached after 6.199s in userspace
sourya@sourya:~$
```

Figure 10: Boot speed after using e4rat

- Access the log file by going to the e4rat folder and using the following command:

```
cd /var/log/e4rat
```

- If you do not find any log file, repeat the above mentioned process. Once the log file is there, access the Grub menu again and press 'e' as your option.
- Enter 'single' at the end of the same line that you have edited before. This will help you access the command line. If a different menu appears asking for anything, choose *Resume normal boot*. If you don't get to the command prompt for some reason, hit *Ctrl+Alt+F1*.
- Enter your details once you see the login prompt.
- Now enter the following command:

```
sudo e4rat-realloc /var/lib/e4rat/
```

```
startup.log
```

This process takes a while, depending on the machine's disk speed.

- Now restart your machine using the following command:

```
sudo shutdown -r now
```

- Now, we need to configure Grub to run e4rat at every boot.
- Access the *grub* file using any editor. For example, *gksu gedit /etc/*

*default/grub*.

- Look for a line starting with *GRUB\_CMDLINE\_LINUX\_DEFAULT=*, and add the following line in between the quotes and before whatever options there are:

```
init=/sbin/e4rat-preload 18
```

- It should look like this:

```
GRUB_CMDLINE_LINUX_DEFAULT = init=/
sbin/e4rat- preload quiet splash
```

- Save and close the Grub menu and update Grub using *sudo update-grub*.

- Reboot the system and you will find noticeable changes in boot speed.

Figures 9 and 10 show the differences between the boot-up time before and after installing e4rat. The improvement in the boot-up speed can be noticed. The time taken to boot before using e4rat was 22.32 seconds, whereas the time taken to boot after using e4rat is 9.065 seconds

## A few simple tweaks

A good boot-up speed can also be achieved using very small tweaks, two of which are listed below.

**SSD:** Using solid-state devices rather than normal hard disks or other storage devices will surely improve your booting speed. SSDs also help in achieving great speeds in transferring files and running applications.

**Disabling GUI:** The graphical user interface, desktop graphics and window animations take up a lot of resources. Disabling the GUI is another good way to achieve great boot-up speed. **END** 🐧

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